

BSDS MATHEMATICS II MIDTERM EXAMINATION FIRST PART

Date: 18th March, 2026

Time: 10:30 am - 11:30 am

Total Marks: 20

- (1) (a) (2+2+1+1 = 6 marks) State the Cauchy-Schwartz Inequality and the Triangle Inequality (and the equivalent conditions for equality in both cases).
(b) (2+2 = 4 marks) Verify the Cauchy Schwartz Inequality and the Triangle Inequality for the vectors $v = (3, -1, 2)^T$ and $w = (1, -1, 1)^T$ using the inner product $\langle v, w \rangle = v^T K w$ where K is the matrix $\begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$.
- (2) (a) (1+1+1+1 = 4 marks) Define the notion of an open set, a closed set, a bounded set and a compact set in $\mathbb{R}^n$.
(b) (2 marks) State the Heine-Borel Theorem.
(c) (2+2 = 4 marks) Consider the set $A = \{(x, y) \in \mathbb{R}^2 \mid 0 < x^2 + y^2 \leq 1\}$. Is A a closed subset of $\mathbb{R}^2$? Is A an open subset of $\mathbb{R}^2$? Briefly justify your answers.

BSDS MATHEMATICS II MIDTER EXAMINATION SECOND PART

Date: 18th March, 2026

Time: 11:45 am - 12:45 pm

Total Marks: 20

- (1) (2+3+5 = 10 marks) What is a positive definite matrix? What is the Cholesky factorization of such a matrix? Find the Cholesky factorization of the following

matrix $\begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}$.

- (2) (2+2+6 = 10 marks) What is an orthogonal matrix? What is the QR decomposition of a nonsingular matrix? Find the QR decomposition of the following matrix

$$\begin{bmatrix} 1 & 4 & 2 \\ 2 & 5 & 3 \\ 3 & 0 & -1 \end{bmatrix}$$